Derivatives and Fixed Income
PGE M1 Fall 2026

Jason Kang

Sessions 4 & 5: Bonds

BUSINESS SCHOOL

SKEMA

Housekeeping



Jason Kang

- Office: B3.09
- Office Hours: E-mail for appointment or walk-in for short queries
- jason.kang@skema.edu

Housekeeping



Scheduling

- 2 sessions on Wednesday Sep. 30 $\rightarrow$ Friday Oct. 15
- 2 sessions on Tuesday Oct. 6 $\rightarrow$ Friday Oct. 9

Plan For Today

- Bond Terminology
- Exercises
- Duration & Convexity

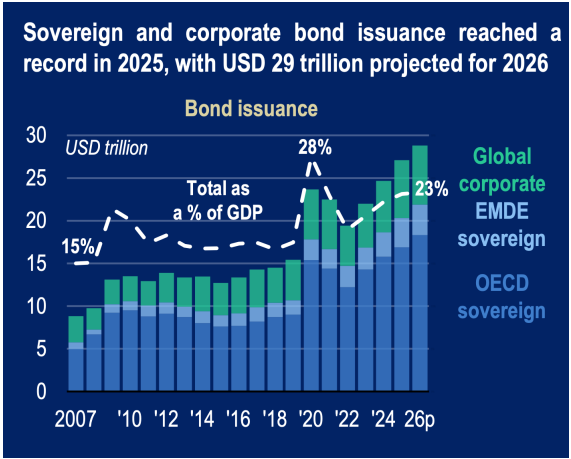
Bonds

A bond is a **debt security** in which the issuer/seller (borrower) raises money from investors today in exchange for a promised repayment at the maturity date.

Example. In July 2026, Amazon raised $25bn through a bond sale to finance its capital expenditures in AI investments. Terms include:

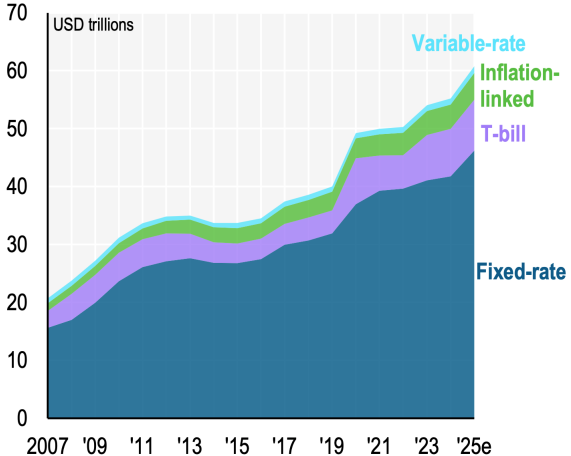
- ▷ Maturities ranging from 3 to 40 years
- ▷ At 65 to 145 basis points (.65-1.45%) above U.S. treasuries
- ⇔ Coupon rates between $4.6 - 6.25\%$

Some Facts: Corporate vs Govt Bonds



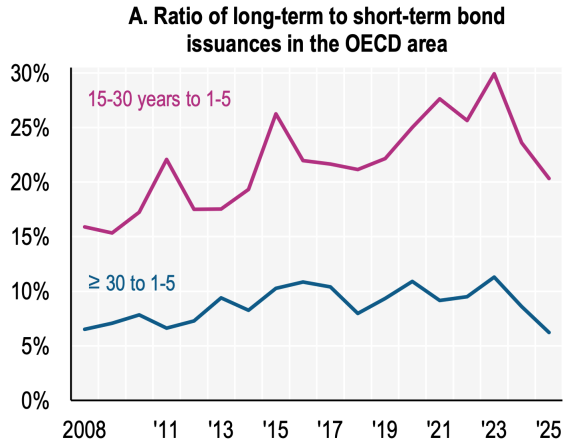
Source: Global Debt Report 2026, *OECD*

Some Facts: Total Issuance of Govt Bonds



Source: Global Debt Report 2026, *OECD*

Some Facts: Maturity of Govt Bonds

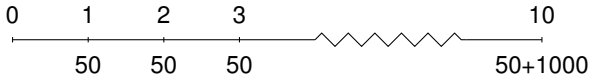


Source: Global Debt Report 2026, *OECD*

Bond Terminology

Bonds typically pay periodic **coupon payments** as a source of intermediate cash flows for investors. At maturity, the issuer repays the principal, or the **face value** of the bond.

Example. Suppose the U.S. issues a 10-year treasury note with a face value of $1000 and a coupon rate of 5%. Then, investors will receive

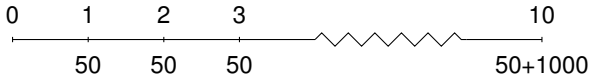


- !! The coupon rate $\neq$ the interest rate on the bond.

Corporations. Investment-grade, junk, callable, convertible bonds.
Governments. Treasury bills, notes, bonds, municipal bonds (local)

Digression

Suppose you are scheduled the receive the below stream of cash flows.



Due to the time value of money, we consider the *present value* of the stream of cash flows as

$$PV_0 = \frac{50}{1+r} + \frac{50}{(1+r)^2} + \cdots + \frac{1050}{(1+r)^{10}}$$

where r is the rate of interest, or the “exchange rate” across different points in time.

Digression

In reality, the cost of borrowing over 1 year should differ from the cost of borrowing over 10 years.

$$PV_0 = \frac{50}{1 + r_1} + \frac{50}{(1 + r_2)^2} + \cdots + \frac{1050}{(1 + r_{10})^{10}}$$

where $r_1, r_2, ..., r_{10}$ are time-specific discount rates.

Here,

r_{10} corresponds to the rate of interest used to discount cash flows that only arrive at $t = 10$ years

Usually, $r_1 < r_2 < r_3 < ... < r_{10}$

Zero-Coupon Bond

A **zero-coupon** bond is a bond that only repays the face value at maturity (there are no coupon payments).

Example. Suppose the U.S. issues a 5-year a zero-coupon bond with face value of $1000. If investors are willing to pay $783.53 for this bond, what is the implied rate of interest?

$$\text{price} = \frac{\text{face value}}{(1+\text{rate})^t}$$

$$783.53 = \frac{1000}{(1+r)^5}$$

$$r = \left(\frac{1000}{783.53}\right)^{1/5} - 1 = 5\%$$

How to interpret? An investor pays $783.53 today to receive $1000 in 5 years. This gives a return of (1000-783.53)/783.53=27.6%, which, on an annual basis, is 5%.

Spot rate

The **spot rate** is the discount rate that equates the present value of a zero-coupon bond to its current market price.

The spot rate computed for a ZCB maturing in N years is referred to as the N-year spot rate.

$*$ From the previous example, we would say that the 5-year spot rate is 5%.

Notation

- $$p_t^{(N)}$$

  - p: price
  - t: reference date (from present value perspective)
  - N: maturity
- The price today of a bond that matures in 2 years: $p_0^{(2)}$
- Sometimes, we need to be more granular:

$$p_{t+1}^{(N \to N-1)} \quad \Leftrightarrow \quad p_1^{(3 \to 2)}$$

price next year of a bond that was a 3 year bond that is now a 2 year bond.

Notation

▷

$$s_t^{(N)}$$

- s: spot rate
- t: reference date (from present value perspective)
- N: maturity

▷ The spot rate today used to discount cash flows that mature in 2 years: $s_0^{(2)}$

▷ For a zero-coupon bond,

$$p_0^{(N)} = \frac{F}{(1+s_0^{(N)})^N} \Rightarrow s_0^{(N)} = \left(\frac{F}{p_0^{(N)}}\right)^{\frac{1}{N}} - 1$$

Spot rate

From the previous example,

$$p_0^{(5)} = 783.53 \quad ; \quad F = 1000$$

Then,

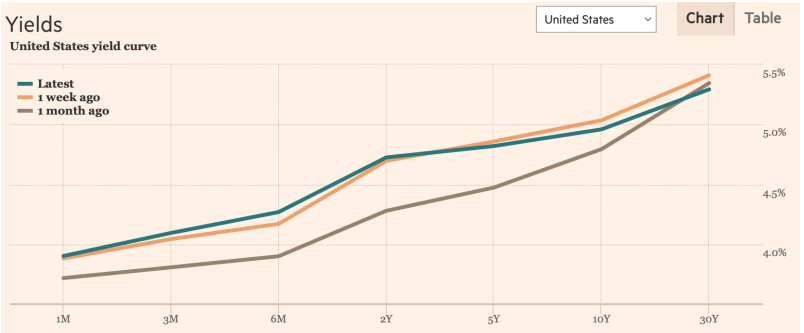
$$s_0^{(N)} = \left(\frac{F}{p_0^{(N)}} \right)^{\frac{1}{N}} - 1 = \left(\frac{1000}{783.53} \right)^{\frac{1}{5}} - 1 = 5\% = s_0^{(5)}$$

Yield Curve

The concept of spot rates apply exclusively to government bonds. For a given market (U.S., Eurozone, Japan, etc) they are considered to be the **risk-free rate** for the corresponding maturity.

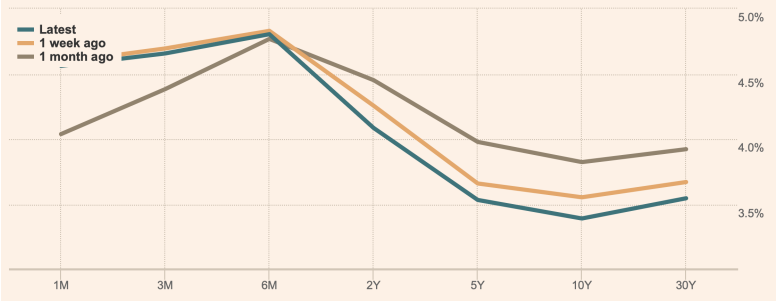
The **yield curve** is a collection of spot rates across bonds of different maturities (by the same issuer).

Yield Curve



Source: Financial Times, as of September 2026

The yield curve (5 years ago)

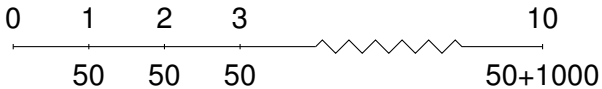


Source: Financial Times, as of January 2022

The shape of the yield curve can change over time!

Coupon-Bearing Bonds

Spot rates allow us to match cash flows that accrue at different points in time to their corresponding discount rates.



Using spot rates,

$$PV_0 = p_0^{(10)} = \frac{50}{1 + s_0^{(1)}} + \frac{50}{(1 + s_0^{(2)})^2} + \cdots + \frac{1050}{(1 + s_0^{(10)})^{10}}$$

Example

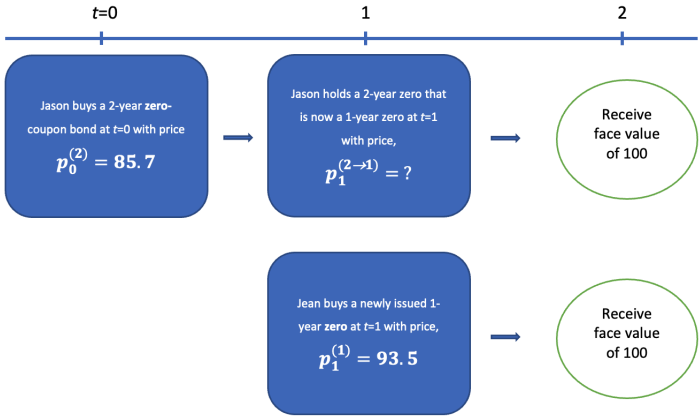


Suppose France issues a 2-year coupon bond with face value of €1000 and a coupon rate of 6% paid annually. What cash flows will you receive?

Suppose the 1-year and 2-year spot rates are 5% and 7% respectively. What should be the price of this bond?

983

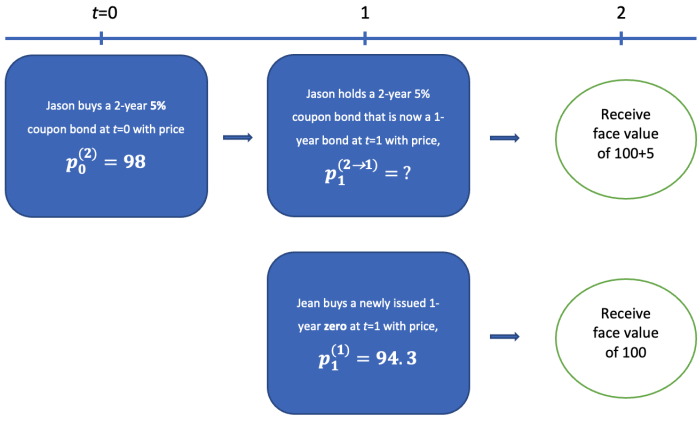
Digression: Arbitrage



What's the price of $p_1^{(1)}$?

By no-arbitrage, $p_1^{(2\to1)} = p_1^{(1)} = 93.5$

Digression: Arbitrage



What's the price of $p_1^{(2\to1)}$? Since $p_1^{(1)}$ is a ZCB, $s_1^{(1)} = \left(\frac{100}{94.3}\right) - 1 = 6\%$.

$$\text{By no-arbitrage, } p_1^{(2\to1)} = \frac{5 + 100}{1 + s_1^{(1)}} = 99.06$$

Digression: Arbitrage

Suppose you conclude the price of the bond should be $p_1^{(2\to 1)} = 99.06$, but the bond is trading in the market at 98. How can you profit? What's the expected return on this strategy?

return=7.14%

Yield-to-Maturity (YTM)

Due to the law of one price (LOOP) and no-arbitrage, the price of a coupon bond should equal

$$p_0^{(N)} = \frac{C}{1 + s_0^{(1)}} + \frac{C}{(1 + s_0^{(2)})^2} + \cdots + \frac{C + F}{(1 + s_0^{(N)})^N}$$

In practice, it is convenient to express the price of a coupon bond using a single discount rate,

$$p_0^{(N)} = \frac{C}{1 + y} + \frac{C}{(1 + y)^2} + \cdots + \frac{C + F}{(1 + y)^N}$$

where y is the **yield-to-maturity** (YTM) of the bond.

Yield-to-Maturity (YTM)

The **yield-to-maturity** of a bond is the unique discount rate that sets the present value of all promised cash flow payments equal to the current market price.

Recall from a previous example, the price today of a 2-year coupon bond with $F = 1000$, $C = 60$ and $s_0^{(1)} = 5\%$, $s_0^{(2)} = 7\%$ is $p_0^{(2)} = 983$.

$$p_0^{(2)} = 983 = \frac{60}{1+y} + \frac{1060}{(1+y)^2} \quad \Rightarrow y = 6.94\%$$

Disclaimer: following our convention, y_0 should denote the YTM of a bond based on prices today. However, some references use y_t to denote spot rates, so y will be used without a time subscript.

Important info re: YTM

i The YTM is the average rate of return of purchasing a bond today and **holding the bond to maturity** while reinvesting intermediate coupon payments.

ii The spot rate for a ZCB is equal to the YTM of that bond.
Recall, a 5-year ZCB with price 783.53 today gives a 5-year return of 27.6%.

iii Memorize the below

When the bond price is	We say the bond trades	This occurs when
greater than the face value	"above par" or "at a premium"	Coupon Rate > Yield to Maturity
equal to the face value	"at par"	Coupon Rate = Yield to Maturity
less than the face value	"below par" or "at a discount"	Coupon Rate < Yield to Maturity

Important info re: YTM

iii (Continued)

- Due to the time-value of money, a ZCB must always trade at a discount.
- A coupon-bearing bond trades at “par” if the coupon rate = yield-to-maturity of that bond.

Important info re: YTM

iv Mechanically, there is an inverse relationship between the bond price and its yield

Important info re: Spot rates

i A downward sloping yield curve predicts an economic contraction in the future. Conversely, an upward sloping yield predicts an economic expansion.

ii On average, given a **coupon-bearing** bond with maturity N, the N-year spot rate $s^{(N)}$ is greater than the YTM on that bond, $s^{(N)} > y$ while $s^{(1)} < y$

Plan For Today



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Exercise 4.10



Suppose that 6-month, 12-month, 18-month, 24-month, and 30-month zero rates are, respectively, 4%, 4.2%, 4.4%, 4.6%, and 4.8% per annum, with continuous compounding. Estimate the cash price of a bond with a face value of 100 that will mature in 30 months and pay a coupon of 4% per annum semiannually.

$\rightarrow$ 98.04

Exercise 4.19



The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

Exercise 4.19



The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

Instrument	**Price**	**Coupon Rate**
6-month T-bill	94.00	0%
1-year T-bill	89.00	0%
1.5-year T-bond	94.84	8%
2-year T-bond	97.12	10%

Exercise 4.19

The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

The first step is usually to find spot rates using ZCB's,

$$p_0^{(.5)} = 94 = \frac{100}{(1 + s_0^{(.5)})^{.5}}$$

$$(1 + s_0^{(.5)})^{.5} = \frac{100}{94}$$

$$s_0^{(.5)} = \left(\frac{100}{94}\right)^2 - 1 = 13.17\%$$

Exercise 4.19

The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

Doing the same for the 1-year ZCB,

$$p_0^{(1)} = 89 = \frac{100}{1 + s_0^{(1)}}$$

$$1 + s_0^{(1)} = \frac{100}{89}$$

$$s_0^{(1)} = \left(\frac{100}{89}\right)^2 - 1 = 12.36\%$$

Exercise 4.19

The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

We have $s_0^{(.5)} = 13.17\%,\ s_0^{(1)} = 12.36\%$.

$$
\begin{aligned}
p^{(1.5)} = 94.84 &= \frac{4}{(1+s^{(.5)})^{.5}} + \frac{4}{1+s^{(1)}} + \frac{104}{(1+s^{(1.5)})^{1.5}} \\
&= \frac{4}{(1.1317)^{.5}} + \frac{4}{1.1236} + \frac{104}{(1+s^{(1.5)})^{1.5}} \\
87.52 &= \frac{104}{(1+s^{(1.5)})^{1.5}} \\
s^{(1.5)} &= \left(\frac{104}{87.52}\right)^{1/1.5} - 1 = 12.19\%
\end{aligned}
$$

Exercise 4.19

The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

Repeat using $s_0^{(.5)} = 13.17\%,\ s_0^{(1)} = 12.36\%,\ s_0^{(1.5)} = 12.19\%$.

$$p^{(2)} = 97.12 = \frac{5}{(1.1317)^{.5}} + \frac{5}{1.1236} + \frac{5}{(1.1219)^{1.5}} + \frac{105}{(1+s^{(2)})^2}$$

$$83.76 = \frac{105}{(1+s^{(2)})^2}$$

$$s^{(2)} = \left(\frac{105}{83.76}\right)^{.5} - 1 = 11.96\%$$

Exercise 4.19

The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of \$4 every 6 months currently sells for \$94.84. A 2-year Treasury bond that will pay coupons of \$5 every 6 months currently sells for \$97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year Treasury zero rates. Assume the face value of these bonds is $F = 100$.

$\rightarrow$We also could have solved via continuous compounding,

$$\begin{aligned} 94 &= 100e^{-s^{(.5)} \times .5} \\ \frac{94}{100} &= e^{-s^{(.5)} \times .5} \\ \log(.94) &= -s^{(.5)} \times .5 \\ s^{(.5)} &= 12.38\% \end{aligned}$$

Since 12.38% is the continuously compounded annual rate, we can convert to the equivalent annual rate which equals $e^{.1238} - 1 = 13.17\%$ as before.

Example

Bond A is a 1-year zero-coupon bond with a yield to maturity of 6%. Bond B is a 2-year, 8% coupon-bond with a yield to maturity of 10%. Bond C is a 2-year zero-coupon bond. All bonds are government bonds with a face value of $\$1,000$. Coupons are paid annually at year-end. Answer the following questions with 4 decimal points for yields and prices:

i. What is the price of Bond A?

ii. What is the price of Bond B?

iii. What is the price of Bond C?
(Hint: write the price of bond B in terms of spot rates.)

Example

Bond A is a 1-year zero-coupon bond with a yield to maturity of 6%. Bond B is a 2-year, 8% coupon-bond with a yield to maturity of 10%. Bond C is a 2-year zero-coupon bond. All bonds are government bonds with a face value of $\$1,000$. Coupons are paid annually at year-end. Answer the following questions with 4 decimal points for yields and prices:

i What is the price of Bond A?

$$p^{(N)} = \frac{F}{(1+y)^N} = \frac{F}{(1+s^{(N)})^N} \Rightarrow p^A = \frac{1000}{1.06} = 943.396$$

ii What is the price of Bond B?

$$p^B = \frac{80}{1.10} + \frac{1080}{1.10^2} = 965.289$$

Here we used the YTM, but we could also represent using spot rates

$$p^B = \frac{80}{1+y} + \frac{1080}{(1+y)^2} \quad ; \quad p^B = \frac{80}{1+s^{(1)}} + \frac{1080}{(1+s^{(2)})^2}$$

Example

Bond A is a 1-year zero-coupon bond with a yield to maturity of 6%. Bond B is a 2-year, 8% coupon-bond with a yield to maturity of 10%. Bond C is a 2-year zero-coupon bond. All bonds are government bonds with a face value of $\$1,000$. Coupons are paid annually at year-end. Answer the following questions with 4 decimal points for yields and prices:

iii. What is the price of Bond C?

We want to know $p^C = p^{(2)} = \dfrac{1000}{(1+s^{(2)})^2}$.

$$\text{Look @ 2-year coupon:} \quad p^B = 965.289 = \frac{80}{1+s^{(1)}} + \frac{1080}{(1+s^{(2)})^2}$$

$$\text{We know } s^{(1)} \text{ from } A: \quad p^B = 965.289 = \frac{80}{1.06} + \frac{1080}{(1+s^{(2)})^2} \quad \Rightarrow s^{(2)} = 10.17\%$$

$$\text{Finally:} \quad p^C = \frac{1000}{(1.1017)^2} = 823.905$$

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Interest Rate Risk: Maturity



- Suppose we have a 5-year ZCB with a face value of 100 and a YTM=5%. What happens if the yield-to-maturity increases by one percentage point to 6%?

$$\frac{100}{1.05^5} = 78.35 \quad \rightarrow \quad \frac{100}{1.06^5} = 74.73$$

Prices fell by 3.63, which is a percentage drop of (74.73-78.35)/78.35= -4.6%.

- What if we were holding a 10-year ZCB?

$$\frac{100}{1.05^{10}} = 61.39 \quad \rightarrow \quad \frac{100}{1.06^{10}} = 55.84$$

which is a percentage drop of (55.84-61.39)/61.39= -9%.

Interest Rate Risk: Coupon Rates



- Suppose we have a 2-year coupon with a face value of 100, a coupon rate of 2% and a YTM=5%. What happens if the yield-to-maturity increases by one percentage point to 6%?

$$\frac{2}{1.05} + \frac{102}{1.05^2} = 94.42 \quad \rightarrow \quad \frac{2}{1.06} + \frac{102}{1.06^2} = 92.67$$

which is a percentage drop of (92.67-94.42)/94.42= -1.9%.

- What if the coupon rate was 5%?

$$100 \quad \rightarrow \quad \frac{5}{1.06} + \frac{105}{1.06^2} = 98.17$$

which is a percentage drop of (98.17-100)/100= -1.8%.

Drivers of Interest Rate Risk

All else equal,

- **Maturity**
  Longer maturity $\rightarrow$ higher interest rate risk

- **Coupon Rate**
  Higher coupon rates $\rightarrow$ lower interest rate risk

Duration

The **duration** of a bond is a measure of its sensitivity to interest rate changes. Quite literally, it measures how many years it takes for a bond to pay back its face value.

Let p denote the price of a bond given maturity N and YTM y. The **Macaulay duration** is given by

$$D_{\text{Macaulay}} = \frac{1}{p} \sum_{n=1}^{N} n \times \frac{C_n}{(1+y)^n}$$

where C_n are all cash flows at time n (including the face value).

Macaulay Duration

Based on

$$D_{\text{Macaulay}} = \frac{1}{p} \sum_{n=1}^{N} n \times \frac{C_n}{(1+y)^n}$$

We can construct a table for a bond with a coupon rate of 5%, face value of 100 and YTM of 6%.

Time (years)	*Cash flow*	*Present value*	*Weight*	*Time* $\times$ *weight*
1	5	4.717	0.049	0.049
2	5	4.450	0.046	0.093
3	5	4.198	0.044	0.131
4	5	3.960	0.041	0.165
5	105	78.463	0.819	4.097
Duration	125	95.787	1.000	4.535

Modified Duration

The **modified duration** is given by

$$D_{\text{modified}} = \frac{1}{1+y} \times D_{\text{Macaulay}}$$

Interpretation. For a bond with a modified duration of N years, a one

percentage point change in the yield leads to approximately N percentage price change in the bond price.

Modified Duration

Recall, a bond with a coupon rate of 5%, face value of 100 and YTM of 6% has a modified duration of 4.465 years and a price of 95.787 today. A one percentage point *decrease* in the YTM from 6% to 5% leads to...

Since the coupon rate = YTM, the bond trades at par and $p = 100$. The percentage price change is

$$\frac{\Delta p}{p} = \frac{100 - 95.787}{95.787} = 4.21\%$$

For comparison,

$$D_{\text{modified}} = \frac{1}{1+y} \times D_{\text{Macaulay}} = \frac{1}{1.06} \times 4.465 = 4.21\%$$

so the interpretation is correct.

First and Second Order Terms

Let p denote the price of a bond, and let $p(y)$ indicate that p is a function of y. For a small change in the ytm,

$$p(y+\Delta y) \approx p(y) + p'(y)\Delta y + \frac{1}{2}p''(y)(\Delta y)^2 + \cdots$$

$$\Delta p \approx p'(y)\Delta y + \frac{1}{2}p''(y)(\Delta y)^2 + \cdots$$

$$\frac{\Delta p}{p} \approx \frac{p'(y)}{p}\Delta y + \frac{1}{2}\frac{p''(y)}{p}(\Delta y)^2 + \cdots$$

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}\frac{p''(y)}{p}(\Delta y)^2 + \cdots$$

When Δy is small, the second-order terms $\to 0$

Convexity

When Δy is not small, we cannot ignore the second-order terms.
Convexity is

$$\text{Convexity} = \frac{1}{p}\frac{d^2p}{dy^2}$$

Based on

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}\frac{p''(y)}{p}(\Delta y)^2 + \cdots$$

we can express the change in prices by

$$\frac{\Delta p}{p} \approx -\text{Duration}_{\text{mod}}\Delta y + \frac{1}{2}\text{Convexity}(\Delta y)^2$$

Convexity

From

$$\text{Convexity} = \frac{1}{p}\frac{d^2p}{dy^2} = \frac{1}{(1+y)^2}\frac{1}{p}\sum_{n=1}^{N} n(n+1)\frac{C_n}{(1+y)^n}$$

and again using a bond with a coupon rate of 5%, face value of 100 and YTM of 6%.

Time n	$n(n+1)$	*Cash flow*	$\frac{n(n+1)\cdot C_n}{(1+y)^2 p(1+y)^n}$
1	2	5	0.087
2	6	5	0.248
3	12	5	0.468
4	20	5	0.736
5	30	105	21.871
Convexity			23.410

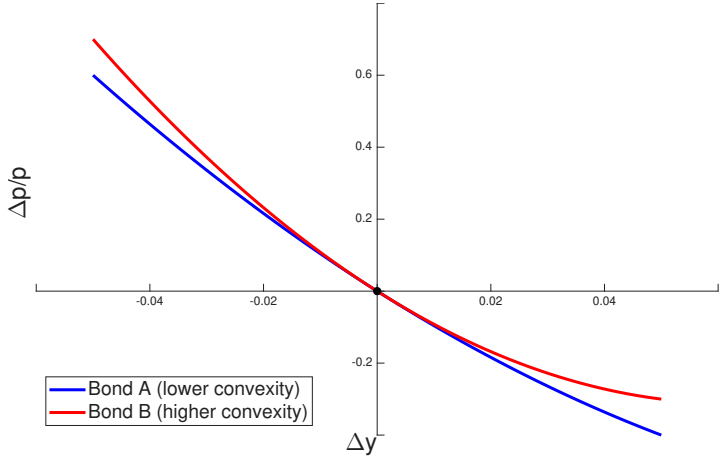
Convexity

Corrections to the percentage price change for given levels of convexity

Δy	$\frac{1}{2}(20)(\Delta y)^2$	$\frac{1}{2}(30)(\Delta y)^2$
.01	+0.10%	+0.15%
.02	+0.40%	+0.60%
.03	+0.90%	+1.35%
.05	+2.50%	+3.75%

High convexity is strictly better for investors.

Convexity



Convexity captures the curvature of the change in prices with respect to the change in yields.

Duration (with continuous compounding)



The Macaulay duration with continuous compounding is given by

$$D_{\text{Macaulay}} = \frac{1}{p} \sum_{n=1}^{N} n \times C_n \times e^{-y \times n}$$

With continuous compounding, we use the Macaulay duration directly.

Exercise 4.18



A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume the face value is $F = 100$.

a. What is the bond's price?

b. What is the bond's duration?

c. Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.

d. Recalculate the bond's price on the basis of a 6.8% per annum yield and verify that the result is in agreement with your answer to (c).

Exercise 4.18

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume the face value is $F = 100$.

a. What is the bond's price?

$$p^{(N)} = \sum_{n=1}^{N} Ce^{-r \times n} + Fe^{r \times n}$$

$$p^{(5)} = \sum_{n=1}^{N} 8e^{-0.07 \times n} + 100e^{0.07 \times 5}$$

$$p^{(5)} = 103.05$$

Exercise 4.18

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume the face value is $F = 100$.

b. What is the bond's duration?

$$D_{\text{Macaulay}} = \frac{1}{p}\sum_{n=1}^{N} n \times C_n \times e^{-y \times n}$$

n	Cash Flow	Discount Factor	$n/p \times$ PV
1	8	0.93239	0.07238
2	8	0.86936	0.13498
3	8	0.81058	0.18878
4	8	0.75578	0.23469
5	108	0.70469	3.69264
Duration			4.32347

Exercise 4.18

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume the face value is $F = 100$.

c. Use the duration to calculate the effect on the bond's price of a 0.2% decrease in its yield.

Interpretation of duration: a Δy percentage point increase in the YTM leads to a $-D\Delta y$ percentage change in the bond price.

$*$This is equivalent to: a Δy percentage point increase in the YTM leads to a $-p \times D\Delta y$ change in prices following

$$\Delta p \approx -p \times \text{Duration} \times \Delta y$$

So, $\Delta p = -103.05 \times 4.3235 \times (-0.002) = 0.89$ increase.
$p = 103.05 + .89 = 103.94$

Exercise 4.18

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume the face value is $F = 100$.

d. Recalculate the bond's price on the basis of a 6.8% per annum yield and verify that the result is in agreement with your answer to (c).

$$p^{(N)} = \sum_{n=1}^{N} Ce^{-r \times n} + Fe^{r \times n}$$

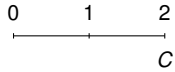
$$p^{(5)} = \sum_{n=1}^{N} 8e^{-0.68 \times n} + 100e^{0.68 \times 5}$$

$$p^{(5)} = 103.94$$

Plan For Today

- Bond Terminology
- Exercises
- Duration & Convexity
- Intro: Dynamics of Bond Prices

Housekeeping



- Suppose we're promised a cash flow at date 2:
  - If rates are variable, its present value is

$$PV_0 = \frac{C}{(1 + r_{0\to 1})(1 + r_{1\to 2})}$$

  - Using spot rates, this is equivalent to

$$PV_0 = \frac{C}{(1 + s_0^{(1)})(1 + \mathbb{E}_0(1 + s_1^{(1)}))}$$

    where $\mathbb{E}_0$ is the expected value at date 0 of the 1-year spot rate in date 1
  - If we instead use the 2-year spot rate,

$$PV_0 = \frac{C}{(1 + s_0^{(2)})^2}$$

    This implies $s_0^{(2)} \approx \sqrt{(1 + s_0^{(1)})(1 + \mathbb{E}_0(1 + s_1^{(1)}))} - 1$